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Tabriz , Iran

 [Nima Falaki](#)

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HIGHLIGHTS

- Electrical Engineer with an M.Sc. in Telecommunication Systems, focused on Quantum Machine Learning and Artificial Intelligence.
- Research experience in hybrid quantum-classical learning, explainable machine learning, anomaly detection, Digital Twin systems, and ML-based modeling and optimization.
- Experienced in developing and evaluating machine-learning pipelines for financial and industrial applications, with emphasis on interpretability, robustness, and practical implementation.

RESEARCH INTERESTS

- Quantum Computing & QML
- Machine Learning
- Explainable & Trustworthy AI
- Digital Twin

EDUCATION

M.Sc. of Telecommunications Systems

Sahand University of Technology
since 2024

- GPA: 3.8/4(18.27/20)
- Supervisor: Dr. Afshin Ebrahimi
- Thesis: An Explainable Quantum-Classical Mixture-of-Experts Framework for Credit Card Fraud Detection

Sahand University of Technology
2019 - 2024

Bachelor of Science in Electrical Engineering

- GPA: 3.59/4 (16.94/20)
- Thesis:
 - Lawn-mowing robot
 - Designing a tunable filter with Photonic Crystal by using the Kerr effect

PUBLICATIONS

- **N. Falaki**, A. Ebrahimi, "Capacity-Aware Credit-Card Fraud Alert Triage with a Safety-Constrained Hybrid Quantum-Classical Mixture-of-Experts and Frozen Real-Hardware Validation," *under review*, 2026.
- **N. Falaki**, A. Ebrahimi, "Personalized and Explainable Financial Transaction Anomaly Detection under Finite Daily Review Capacity," *Scientific Reports*, *under review*, 2026.
- **N. Falaki**, A. Ebrahimi, "P-MVARS: Personalized Multi-View and Capacity-Aware Anomaly Ranking in Digital Financial Transactions," *12th International Symposium on Telecommunication (IST'2026)*, 2026.
- **N. Falaki**, A. Eteghadi, A. Panahgholi, A. Ebrahimi, "A Data-Driven Framework for Demand Forecasting and Inventory Optimization in ERP Using Machine Learning and DDMRP on FPGA," *2026 Second International Biennial Conference on Advances in Artificial Intelligence and Data Science (IBAAIDS)*, 2026.
[2026.doi.org/10.1109/IBAAIDS66771.2026.11567409](https://doi.org/10.1109/IBAAIDS66771.2026.11567409)
- F. Barzgar, A. Alimohammadi, **N. Falaki**, A. Ebrahimi, H. Azmi, "A Simulation-Based Digital Twin with Transition-Aware Energy Forecasting for Smart Building Energy Management," *11th International Conference on Energy Technology and Management*, 2026.
- N. Esmaeili, A. Ebrahimi, F. Ekhtiari Kashki, **N. Falaki**, "Application of Artificial Intelligence in Condition Monitoring of Steam Turbine Oil in Power Plants (in Persian) " *36rd Iranian Conference on Electrical Engineering (ICEE)*.

PROJECTS

Maadi AI Language Tutor — AI-Powered Language Learning Chatbot

Maadi English Academy

2026

- Designed and developed an AI-powered language learning chatbot for teaching support, language practice, and student feedback.

Smart Interactive Robot

Sahand University of Technology

2024

- Launching with Raspberry Pi
- Implementing facial recognition using neural network models and object detection with Yolov model
- voice detection and responsiveness features

Implementing a GPS System for Vehicle Tracking

Sahand University of Technology

2023

- Launching with Node MCU , Neo-6 and Sim card module Setting up Wi-Fi for sending data to the server Implementing SMS for data transmission

WORK EXPERIENCE

Instructor

Yasan Academy

2023 - Present

- Teaching Python at both basic and advanced levels
- Teaching Web Design
- Teaching Internet of Things with NodeMCU board

Internship

Sahand University of Technology

July 2022 - September2022

Linear Control Lab for Electronic System Fault Diagnosis

SKILLS

- Quantum Computing and Quantum Machine Learning
- Machine Learning, Deep Learning, and Explainable AI
- Digital Twin and Data-Driven Modeling
- Signal Processing
- Python and MATLAB
- FPGA/Embedded Systems and Hardware-Software Integration

HONORS AND AWARDS

- Member, Iran’s National Elites Foundation — 2026
- Member, Office of Exceptional Talents, Sahand University of Technology — 2025–2026
- Ranked 1st in M.Sc. Telecommunication Systems, Sahand University of Technology — 2025–2026
- Ranked 2nd in B.Sc. Electronics, Sahand University of Technology — 2024
- Exceptional Talent Admission to the M.Sc. Program, Sahand University of Technology — Admitted without the national graduate entrance examination

VOLUNTEER EXPERIENCE

- Vice Chair, IEEE Student Branch, Sahand University of Technology — 2025–2026
- Student Committee Member, IEEE Iran Section — 2024–2026
- Organizer, IntelliTech AI Event — 2025
- Member, Electrical Engineering Scientific Association, Sahand University of Technology — 2022–2023
- Member of IEEE Young Professionals Iran section

LANGUAGES

Persian: Native | Azerbaijani: Native | Turkish: Native
English: Fluent | IELTS Academic booked for 19 February 2027

• References, Further information, and Proofs are available upon Request